Model Notes

Endogenous Education Choice in Segmented Search Markets

(Skill-biased P_S, aggregate scale A, discrete cross-market search)

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Scope. This document contains the full mathematical description of the model: primitives, value functions, equilibrium conditions, solution algorithms, and transition dynamics. It serves as the companion reference for the paper and the codebase.

Contents

1 Global environment

Time is continuous. All agents discount at rate $r > 0$.

1.1 Aggregate scale and sectoral productivity shifters

A single aggregate productivity scale $A > 0$ multiplies output in both sectors; it sets the overall dollar level of production and wages and carries no cross-sectional shape information. Net of A, there are two sector-specific productivity shifters $P_U > 0$ and $P_S(\cdot)$. Whereas P_U is a constant scalar, the skilled productivity shifter is an *increasing* function of worker type x, parameterised as the density of a one-parameter Beta distribution:

$$P_S(x) \;=\; \gamma_{P_S}\, x^{\gamma_{P_S}-1}, \qquad x \in [0,1], \qquad \gamma_{P_S} \geq 1, \tag{1}$$

i.e. $P_S(\cdot) \sim \text{Beta}(\gamma_{P_S}, 1)$. The single parameter γ_{P_S} governs the skill gradient. Setting $\gamma_{P_S} = 1$ recovers a flat $P_S(x) = 1$ (skill-neutral); $\gamma_{P_S} > 1$ generates a strict skill premium. Note that γ_{P_S} is simultaneously the skilled coefficient and the exponent in $x^{\gamma_{P_S}}$; this dual role is what separates A (overall level) from the relative skilled productivity in estimation.

1.2 Demographic turnover

Workers exit the economy at Poisson rate $\nu > 0$ and are replaced one-for-one by newborn untrained workers with density $\ell(x)$. This delivers a stationary population composition.

1.3 Stationarity

The model admits a stationary equilibrium (steady state) in which the cross-sectional distribution of workers across states is time-invariant. The transition-dynamics section (Section 12) studies the path generated by an unexpected, permanent change in some parameters at a date T.

2 Parameter classification

Parameters fall into four groups, following the multi-regime estimation strategy described in the paper plan. The classification identifies which parameters are externally set, which are estimated once on the baseline window and held fixed, and which are re-estimated within each window. All parameters appear without a regime argument in the equations below.

2.1 Calibrated / fixed

Symbol	Name	Source / calibration
r	Discount rate	5% annual real rate (risk-free rate)

The discount rate is set externally and held fixed throughout all estimation stages.

2.2 Pre-estimated from data

Symbol	Name	Source / method
ϕ	Training completion rate	NSC/IPEDS Universe enrollment + NCES median time-to-degree, enrollment-weighted across 4-year and 2-year programmes
ν	Demographic turnover rate	CPS Basic Monthly LF→NILF exit hazard (ages 16–64) over the pre-FC baseline window

Both rates are pre-estimated from external micro-data outside the SMM objective. The training completion rate is computed as

$$\hat{\phi} \;=\; \frac{E_{4\text{yr}}/d_{4\text{yr}} + E_{2\text{yr}}/d_{2\text{yr}}}{E_{4\text{yr}} + E_{2\text{yr}}},$$

where E_k is the average IPEDS Universe enrollment in k-year institutions over the sample and d_k is the corresponding NCES median time-to-degree (49 months for 4-year bachelor's; 37 months for 2-year associate's). The turnover rate $\hat{\nu}$ equals the average monthly LF→NILF exit rate for the working-age population (16–64) over the pre-FC baseline window.

2.3 Deep structural parameters

Symbol	Name	Interpretation
a_ℓ, b_ℓ	Type distribution (Beta)	Worker heterogeneity
$\tilde{b}_U$	Unskilled outside flow (rel. to A)	Institutional UBI and UI policy
$\tilde{b}_T$	Training flow (rel. to A)	Training stipend / support
$\tilde{b}_S$	Skilled outside flow (rel. to A)	Institutional UBI and UI policy

These parameters govern the *fundamentals* of the labour market that we assume are invariant across crisis episodes.

2.4 Regime-specific parameters (re-estimated within each window)

Symbol	Name	Interpretation
c	Training cost coefficient	Cost of education
A	Aggregate production scale	Overall TFP / demand level
P_U, γ_{P_S}	Productivity shifters	Relative sectoral productivity
$\tilde{k}_U, \tilde{k}_S$	Vacancy costs (rel. to A)	Hiring conditions
λ_U, λ_S	Shock arrival rates	Match-destruction risk
ξ_S	Exogenous skilled separation	Skilled layoff / dissolution hazard
α_U	Unskilled damage shape	Damage distribution
a_Γ, b_Γ	Skilled offer shape	Offer / quality distribution
η_U, η_S	Matching elasticities	Matching-function curvature
μ_U, μ_S	Matching efficiencies	Matching-function level
β_U, β_S	Bargaining weights	Wage-setting institutions
σ_S	OJS flow cost	On-the-job search cost
$\sigma_{w,U}, \sigma_{w,S}$	Wage measurement error	Survey log-wage noise (per sector)

These parameters capture the *aggregate state* of the economy (A and the relative productivity shifters), the *matching and wage-setting technology*, the *training technology*, and the survey *measurement error* on wages. The pair $(\sigma_{w,U}, \sigma_{w,S})$ is an estimation/measurement device: it enters only the observed wage moments (Section 9), not the equilibrium.

3 Workers, firms, and types

3.1 Workers

Each worker has an innate type $x \in \mathcal{X} = [0, 1]$, fixed over time. Types are distributed according to CDF L with density ℓ ($x \sim \text{Beta}(a_\ell, b_\ell)$).

Market access via training. Workers can participate in either the *unskilled* market U (untrained status) or the *skilled* market S (trained status). Training changes *market access only*; it does not change innate type x.

Training decision at unskilled unemployment. Training can be initiated *only* by untrained workers who are in unskilled unemployment. The optimal policy is characterised by a cutoff $\bar{x} \in [0, 1]$: upon entering unskilled unemployment, types $x > \bar{x}$ initiate training, while types $x \leq \bar{x}$ remain in unskilled search.

Cross-market search by skilled unemployed. A trained worker who is in skilled unemployment chooses, as a function of x, which market to search in:

$$d(x) \in \{0, 1\}, \qquad d(x) = 1 \iff \text{the worker searches the unskilled market.}$$

Searching is exclusive: at each instant the worker is contacted by firms in one market only, at rate $\kappa_S = \theta_S q_S(\theta_S)$ if $d(x) = 0$, and at rate $f_U = \theta_U q_U(\theta_U)$ if $d(x) = 1$. We characterise the optimal $d(x)$ as a comparison of two candidate unemployment values in Section 8. Skilled-employed workers do not search the unskilled market.

Loss of training upon cross-market matching. A skilled-unemployed worker of type x who matches in the unskilled market is treated as *untrained* from the moment of acceptance on: the bargained wage uses $U_U(x)$ as the worker's outside option, the firm receives the standard $J_U(x, 1)$, and if the worker later separates from this match, she re-enters untrained unemployment $u_U(x)$. In particular, she may re-initiate training (paying $c(x)$) at a future date if her type warrants it. We refer to this as the "training is obsolete" interpretation of the cross-market move.

3.2 Firms

There are two labour markets. Both markets have homogeneous firms.

4 Training technology

Eligibility. Only untrained workers in unskilled unemployment may initiate training.

Cost. Initiating training requires a one-time cost $c(x)$ with $c'(x) < 0$. In applications,

$$c(x) = (1 - x)\, e^{(c-x)}.$$

Higher-ability workers face lower training costs. The parameter c enters through exponentiation so that the optimiser searches over $c \in \mathbb{R}$ while the cost coefficient $e^c > 0$ is guaranteed positive.

Completion. Training completes at Poisson rate $\phi > 0$. Upon completion, the worker becomes trained and enters skilled unemployment.

Flow payoffs. Let b_U, b_T, and b_S denote flow payoffs while unskilled unemployed, in training, and skilled unemployed, respectively. A skilled-unemployed worker receives the flow b_S regardless of the search-market choice $d(x)$.

Demographic turnover (stationarity device). Workers exit the economy at Poisson rate $\nu > 0$ and are replaced one-for-one by newborn untrained workers with density $\ell(x)$.

5 Production technologies

Both production functions carry the common aggregate scale $A > 0$. The institutional flow values and the vacancy costs are expressed relative to this scale: $b_j = \tilde{b}_j A$ for $j \in \{U, T, S\}$ and $k_j = \tilde{k}_j A$ for $j \in \{U, S\}$, where the estimated objects are the dimensionless $\tilde{b}_j$ and $\tilde{k}_j$. The value functions and free-entry conditions below are written in the dimensional b_j, k_j.

Unskilled production. An unskilled match between worker type x and a homogeneous firm with match-specific efficiency $p \in (0, 1]$ produces

$$\pi_U(x, p) = A\, P_U\, x\, p.$$

Skilled production. A skilled match between worker type x and a homogeneous firm with idiosyncratic quality $p \in (0, 1]$ produces

$$\pi_S(x, p) \;=\; A\, x\, P_S(x)\, p \;=\; A\, \gamma_{P_S}\, x^{\gamma_{P_S}}\, p, \tag{2}$$

$P_S(x) = \gamma_{P_S}\, x^{(\gamma_{P_S}-1)}$, strictly increasing in x when $\gamma_{P_S} > 1$.

6 Unskilled labour market

6.1 Matching

Let $\mathcal{U}_U$ and $\mathcal{V}_U$ denote the masses of unskilled unemployed and unskilled vacancies. The relevant *seeker* mass for matching is the augmented pool

$$\widetilde{\mathcal{U}}_U \;\equiv\; \int_0^1 \big(u_U(x) + d(x)\, u_S(x)\big)\, \mathrm{d}x \;=\; \mathcal{U}_U + \int_0^1 d(x)\, u_S(x)\, \mathrm{d}x,$$

which adds the skilled-unemployed workers who have chosen to search unskilled. Market tightness on the U-side is

$$\theta_U = \frac{\mathcal{V}_U}{\widetilde{\mathcal{U}}_U}.$$

Matching follows a Cobb–Douglas function with parameters $\mu_U > 0$ and $\eta_U \in (0, 1)$:

$$M_U(\widetilde{\mathcal{U}}_U, \mathcal{V}_U) = \mu_U\, \widetilde{\mathcal{U}}_U^{\eta_U}\, \mathcal{V}_U^{1-\eta_U},$$

implying vacancy filling and job-finding rates

$$q_U(\theta_U) = \mu_U\, \theta_U^{-\eta_U}, \qquad f_U \;\equiv\; \theta_U\, q_U(\theta_U) = \mu_U\, \theta_U^{1-\eta_U}.$$

A contacted worker is of type x with probability $\big(u_U(x) + d(x) u_S(x)\big)/\widetilde{\mathcal{U}}_U$.

6.2 Match productivity and shocks

New matches are created at frontier efficiency $p_0 = 1$. Idiosyncratic damage shocks arrive at Poisson rate $\lambda_U > 0$. Upon a shock, post-shock efficiency is

$$p' = e^{-d}, \qquad d \sim \mathrm{Exp}(\alpha_U),$$

equivalently

$$p' \sim \mathrm{Beta}(\alpha_U, 1), \qquad G(p) = p^{\alpha_U}, \quad p \in [0, 1].$$

A match is endogenously destroyed whenever $p' < p^*(x)$ for a reservation threshold $p^*(x) \in (0, 1)$.

6.3 Value functions and training policy

Let:

- $U_U(x)$ be the value of untrained unskilled unemployment,
- $U_U^{\text{search}}(x)$ the value of unskilled search,
- $T(x)$ the value of training,
- $E_U(x,p)$ the worker value of unskilled employment at efficiency p,
- $J_U(x,p)$ the firm value of a filled unskilled job,
- V_U the value of an unskilled vacancy.

Household values and training policy. An untrained unemployed worker chooses between unskilled search and initiating training:

$$U_U(x) = \max\{U_U^{\text{search}}(x),\ U_U^{\text{train}}(x)\}, \qquad U_U^{\text{train}}(x) := -c(x) + T(x).$$

Training completes at Poisson rate ϕ and leads to skilled unemployment. The training value satisfies

$$(r+\phi+\nu)\,T(x) = b_T + \phi\, U_S(x). \tag{3}$$

Unskilled search yields flow payoff b_U and contacts vacancies at rate $f_U = \theta_U q_U(\theta_U)$:

$$(r+\nu)\,U_U^{\text{search}}(x) = b_U + f_U\big(E_U(x,1) - U_U^{\text{search}}(x)\big). \tag{4}$$

Define the net training gain

$$\Delta_T(x) := -c(x) + T(x) - U_U^{\text{search}}(x).$$

Under monotone incentives, there exists a cutoff $\bar{x} \in [0,1]$ such that

$$\tau(x) = \mathbf{1}\{x > \bar{x}\}, \qquad U_U^{\text{search}}(\bar{x}) = -c(\bar{x}) + T(\bar{x}).$$

6.4 Match values, bargaining, and endogenous separation

A new unskilled match starts at frontier efficiency $p_0 = 1$. Idiosyncratic damage shocks arrive at rate λ_U and redraw efficiency p' from G on $[0,1]$. For $p \in (0,1]$, worker and firm values satisfy

$$\begin{aligned}(r+\nu+\lambda_U)\,E_U(x,p) &= w_U(x,p) \\ &\quad + \lambda_U\left[U_U(x)\,G(p^*(x)) + \int_{p^*(x)}^{1} E_U(x,p')\,\mathrm{d}G(p')\right],\end{aligned} \tag{5}$$

$$(r+\nu+\lambda_U)\,J_U(x,p) = A\,P_U\,x\,p - w_U(x,p) + \lambda_U \int_{p^*(x)}^{1} J_U(x,p')\,\mathrm{d}G(p'). \tag{6}$$

Wages are determined by Nash bargaining with worker weight $\beta_U \in (0,1)$:

$$\beta_U\, J_U(x,p) = (1-\beta_U)\big(E_U(x,p) - U_U(x)\big). \tag{7}$$

Bargaining for skilled-origin matches. A skilled-unemployed worker who matches in the U-market is treated as untrained at the moment of acceptance: her outside option for the Nash split is $U_U(x)$, not $U_S(x)$. Hence the firm value of such a match is the same $J_U(x,1) = (1-\beta_U)\,S_U(x,1)$ that applies to a fresh untrained match. This is the single bookkeeping assumption that lets the unskilled-side free entry use a *single* firm-value surface. Endogenous separation is characterised by a reservation threshold $p^*(x) \in (0,1)$ satisfying

$$J_U(x,p^*(x)) = 0, \qquad J_U(x,p) \geq 0 \iff p \geq p^*(x). \tag{8}$$

6.5 Pointwise update for unskilled search values

Rearranging (4) gives

$$(r + \nu + f_U)\, U_U^{\text{search}}(x) = b_U + f_U\, E_U(x, 1). \tag{9}$$

Using Nash bargaining at entry efficiency $p_0 = 1$,

$$E_U(x, 1) = U_U(x) + \frac{\beta_U}{1 - \beta_U}\, J_U(x, 1),$$

we obtain the pointwise update

$$U_U^{\text{search}}(x) = \frac{b_U + f_U\left(U_U(x) + \frac{\beta_U}{1-\beta_U}\, J_U(x, 1)\right)}{r + \nu + f_U}. \tag{10}$$

Under the cutoff policy $\tau(x) = \mathbf{1}\{x > \bar{x}\}$,

$$U_U(x) = \begin{cases} U_U^{\text{search}}(x), & x \le \bar{x}, \\ U_U^{\text{train}}(x), & x > \bar{x}, \end{cases}$$

so (10) becomes a simple piecewise update. On the search region $x \le \bar{x}$ it reduces to

$$(r + \nu)\, U_U^{\text{search}}(x) = b_U + \frac{\beta_U}{1 - \beta_U}\, f_U\, J_U(x, 1).$$

6.6 Unskilled match surplus (wage-free equation)

Define the total match surplus as worker net gain plus firm value:

$$S_U(x, p) \;\equiv\; \big(E_U(x, p) - U_U(x)\big) + J_U(x, p).$$

Equivalently, $E_U(x, p) + J_U(x, p) = S_U(x, p) + U_U(x)$. Adding the worker and firm HJBs (5)–(6) and substituting $E_U + J_U = S_U + U_U$:

$$(r + \nu + \lambda_U)\, S_U(x, p) = A\, P_U\, x\, p - (r + \nu)\, U_U(x) + \lambda_U \int_{p^*(x)}^{1} S_U(x, p')\, \mathrm{d}G(p'). \tag{11}$$

6.7 Wage schedule (unskilled)

Given the surplus split $J_U = (1 - \beta_U) S_U$ and $E_U - U_U = \beta_U S_U$, wages are recovered from the firm HJB (6). Multiplying the surplus equation (11) by $(1 - \beta_U)$ and comparing with the firm HJB gives

$$w_U(x, p) = \beta_U\, A\, P_U\, x\, p + (1 - \beta_U)(r + \nu)\, U_U(x). \tag{12}$$

The wage is linear in match quality p: the worker captures share β_U of flow output $A\, P_U\, x\, p$, plus a flow option value $(1 - \beta_U)(r + \nu) U_U(x)$ reflecting the firm's outside value when the worker exits (at rate ν) or the match is destroyed. Skilled-detoured workers and original untrained workers are paid the same wage at the same (x, p), by the bargaining assumption above.

6.8 Vacancies, free entry, and market tightness

Let V_U denote the value of an unskilled vacancy. Posting a vacancy costs k_U per unit time and yields contacts at rate $q_U(\theta_U)$. A vacancy meets type x with probability $\big(u_U(x) + d(x) u_S(x)\big)/\widetilde{\mathcal{U}}_U$ and creates a match at $p_0 = 1$, yielding value $J_U(x, 1)$ regardless of seeker origin. The vacancy HJB is

$$rV_U = -k_U + q_U(\theta_U) \left(\frac{\int_0^1 J_U(x, 1)\, \big(u_U(x) + d(x) u_S(x)\big)\, \mathrm{d}x}{\widetilde{\mathcal{U}}_U} - V_U \right). \tag{13}$$

Free entry implies $V_U = 0$, hence

$$q_U(\theta_U) = \frac{k_U\, \widetilde{\mathcal{U}}_U}{\int_0^1 J_U(x,1)\,\big(u_U(x) + d(x) u_S(x)\big)\,\mathrm{d}x}, \tag{14}$$

which pins down market tightness θ_U given the endogenous composition of unemployed workers *including* the skilled-unemployed who have chosen unskilled search. Because A multiplies $J_U(x,1)$ through the surplus, a higher A raises the filled-job value and hence (other things equal) tightness.

6.9 Reservation rule

Setting $S_U(x,p^*) = 0$ in the surplus equation (11):

$$p^*(x) = \frac{(r+\nu)\,U_U(x) - \lambda_U \int_{p^*}^1 S_U(x,p')\,\mathrm{d}G(p')}{A\,P_U\,x}. \tag{15}$$

6.10 Stationary cross-sectional distributions (unskilled)

The cross-sectional distribution of *untrained* workers across states is pinned down by flow balance. Let $u_U(x)$ denote the density of untrained workers in unskilled unemployment, $t(x)$ the density of workers in training, and define the untrained-segment mass by type:

$$m_U(x) \;\equiv\; \ell(x) - m_S(x),$$

where $m_S(x)$ is given by (18) below. The untrained employed density is the residual:

$$e_U(x) \;:=\; m_U(x) - u_U(x) - t(x).$$

This residual now blends original-untrained and skilled-detoured workers, who are indistinguishable in e_U by the bookkeeping choice above.

Training. At equilibrium the training distribution is stationary:

$$0 = \tau(x)\,u_U(x) - (\phi+\nu)\,t(x), \quad\implies\quad t(x) = \frac{\tau(x)}{\phi+\nu}\,u_U(x). \tag{16}$$

Untrained unemployment. Untrained unemployment receives inflow from demographic replacement and from job destruction out of unskilled employment. It loses mass due to job finding, training entry, and demographic exit. The density flow balance is

$$(f_U + \tau(x) + \nu)\,u_U(x) = \nu\,\ell(x) + \lambda_U\,G(p^*(x))\,e_U(x). \tag{17}$$

Separations from skilled-detoured workers contribute to the right-hand side automatically because they enter $e_U(x)$ on the same footing as the original untrained employed.

7 Unskilled-block equilibrium and solution algorithm

An *unskilled-block equilibrium* is a collection

$$\{\theta_U,\ U_U^{\text{search}}(\cdot),\ T(\cdot),\ U_U(\cdot),\ E_U(\cdot,\cdot),\ J_U(\cdot,\cdot),\ p^*(\cdot),\ u_U(\cdot),\ t(\cdot)\}$$

such that:

(E1) **Household values and training.** $T(x)$ satisfies (3); $U_U^{\text{search}}(x)$ satisfies (4) (equivalently (10)); and unemployment values satisfy $U_U(x) = \max\{U_U^{\text{search}}(x),\ -c(x) + T(x)\}$, $\tau(x) = \mathbf{1}\{-c(x) + T(x) \geq U_U^{\text{search}}(x)\}$.

(E2) **Match values and wages.** Worker and firm match values satisfy (5)–(6); wages satisfy Nash bargaining (7). (Equivalently, $w_U(x,p)$ is recovered from (12).)

(E3) **Endogenous separation.** The reservation threshold $p^*(x)$ satisfies (8).

(E4) **Vacancy creation (free entry).** Market tightness θ_U satisfies free entry (14) (augmented seeker pool).

(E5) **Stationary cross-sectional distributions.** Densities satisfy the flow-balance conditions (16) and (17). Employment of untrained workers is the residual mass $e_U(x) = m_U(x) - u_U(x) - t(x)$, where $m_U(x) = \ell(x) - m_S(x)$.

7.1 Solution algorithm (unskilled side)

For computation, the unskilled block is organised as a two-layer fixed point. The *inner loop* solves the value functions given conjectures for market tightness and the reservation rule. The *outer loop* updates the policy objects $(\tau(\cdot), p^*(\cdot), \theta_U)$ and the stationary composition until all equilibrium conditions hold jointly. The cross-market pool $d(\cdot)\, u_S(\cdot)$ is supplied by the skilled block via the global iteration and treated as exogenous within one unskilled block solve.

Outer-loop unknowns. We iterate on

$$\left(\theta_U^{(n)},\ p^{*,(n)}(\cdot),\ \tau^{(n)}(\cdot),\ u_U^{(n)}(\cdot),\ t^{(n)}(\cdot)\right),$$

with $d(\cdot)\, u_S(\cdot)$ held fixed from the previous global pass.

Step 1. Initialise (outer loop). Choose initial guesses $\theta_U^{(0)} > 0$ and a feasible reservation rule $p^{*,(0)}(x) \in (0,1)$. Initialise a training policy $\tau^{(0)}(x)$ and stationary densities $u_U^{(0)}(x)$ and $t^{(0)}(x)$.

Step 2. Inner value-function iteration. Holding $(\theta_U^{(n)}, p^{*,(n)}(\cdot))$ fixed, iterate jointly on $(U_U^{\text{search}}, U_U, T, E_U, J_U)$ until convergence. One convenient inner cycle is:

- **Step 2.1. Training value.** Update $T^{(m+1)}(x)$ from (3), treating $U_S(\cdot)$ as given by the skilled block.
- **Step 2.2. Unemployment value and training decision.** Given $T^{(m+1)}$ and the current search value $U_U^{\text{search},(m)}$, update

$$U_U^{(m+1)}(x) = \max\{U_U^{\text{search},(m)}(x),\ -c(x) + T^{(m+1)}(x)\}$$

  and the implied training policy

$$\tau^{(m+1)}(x) = \mathbf{1}\{-c(x) + T^{(m+1)}(x) \geq U_U^{\text{search},(m)}(x)\}.$$

- **Step 2.3. Match values.** Given $U_U^{(m+1)}(x)$ and the fixed reservation rule $p^{*,(n)}(\cdot)$, solve the coupled system (5)–(6) together with bargaining (7) to obtain $E_U^{(m+1)}(x,p)$ and $J_U^{(m+1)}(x,p)$.
- **Step 2.4. Search value.** Update $U_U^{\text{search},(m+1)}(x)$ from the HJB (4):

$$(r+\nu)\, U_U^{\text{search},(m+1)}(x) = b_U + f_U^{(n)}\,(E_U^{(m+1)}(x,1) - U_U^{\text{search},(m+1)}(x)),$$

  with $f_U^{(n)} = \theta_U^{(n)} q_U(\theta_U^{(n)})$.

Stop the inner loop when the sup norm of updates to $(U_U^{\text{search}}, U_U, E_U, J_U)$ is below tolerance.

Step 3. Update stationary distributions. Using the converged training policy $\tau^{(n+1)}(\cdot)$ from the inner loop and the current reservation rule $p^{*,(n)}(\cdot)$, update the stationary densities $(u_U^{(n+1)}, t^{(n+1)})$ by solving (16)–(17) pointwise in x.

Step 4. Update reservation rule p^* (outer update). Setting $S_U(x, p^*) = 0$ in the closed-form surplus expression yields the direct formula (15). Anderson acceleration (rank-1) is applied to the vector $\{p^{*,(n+1)}(x)\}_x$ before writing it back to cache.

Step 5. Update tightness θ_U by free entry. Given the updated composition $u_U^{(n+1)}(\cdot)$, the carried $d(\cdot)\, u_S(\cdot)$ from the previous global pass, and the converged frontier surplus $J_U(x, 1)$, update θ_U using free entry (14):

$$q_U(\theta_U^{(n+1)}) = \frac{k_U\,(\mathcal{U}_U^{(n+1)} + \int d\, u_S\, \mathrm{d}x)}{\int_0^1 J_U(x,1)\,(u_U^{(n+1)}(x) + d(x)u_S(x))\,\mathrm{d}x},$$

and invert $q_U(\theta) = \mu_U\, \theta^{-\eta_U}$ to obtain $\theta_U^{(n+1)}$. Use Anderson acceleration when updating tightness, to reduce fluctuations.

Step 6. Convergence (outer loop). Stop if

$$\max\Big\{|\theta_U^{(n+1)} - \theta_U^{(n)}|,\ \|p^{*,(n+1)}(\cdot) - p^{*,(n)}(\cdot)\|_\infty, \|u_U^{(n+1)}(\cdot) - u_U^{(n)}(\cdot)\|_\infty\Big\} < \text{tol}.$$

Otherwise set $n \leftarrow n + 1$ and return to Step 2.

Remark 1 (Two-block model)**.** *In the full model, Step 2 takes $U_S(\cdot)$ as given from the skilled block, while the skilled block takes the trained mass $m_S(\cdot)$ (see (18) below) as given from the unskilled block. In addition, the unskilled free entry consumes $d(\cdot)\, u_S(\cdot)$ from the skilled block, and the skilled inner loop consumes $E_U(\cdot, 1)$ and f_U from the unskilled block. A global equilibrium is obtained by iterating on the two blocks until these link variables are consistent.*

8 Skilled labour market

This section adapts the Pissarides (2000) Chapter 4 framework ("Labour Turnover and On-the-Job Search") to our segmented environment with worker heterogeneity $x \in [0, 1]$. The skilled block features (i) idiosyncratic match quality $p \in [0, 1]$ that is redrawn by shocks, (ii) endogenous separation via a cutoff $p_S^*(x)$, (iii) on-the-job search (OJS) as a discrete choice with a flow search cost σ_S, and (iv) the discrete *search- market choice* $d(x) \in \{0, 1\}$ for skilled-unemployed workers. Skilled separations arise from two channels: an exogenous layoff hazard $\xi_S \geq 0$ and the endogenous mechanism (a quality shock that lands below $p_S^*(x)$).

8.1 Primitives and market objects

Trained population. Only trained workers participate in the skilled market. Let $m_S(x)$ be the density of trained workers of type x. In stationarity — and incorporating the cross-market outflow —

$$m_S(x) \;=\; \frac{\phi\, t(x)}{\nu + d(x)\, f_U}. \tag{18}$$

For $d(x) = 0$ types this reduces to $m_S(x) = (\phi/\nu)\, t(x)$; for $d(x) = 1$ types the extra outflow f_U adds to the denominator. Exogenous separation ξ_S moves workers from skilled employment to skilled unemployment *within* the segment, so it does not affect $m_S(x)$.

Production. A skilled match between worker type x and idiosyncratic quality $p \in [0,1]$ produces $\pi_S(x,p) = A\,\gamma_{P_S}\,x^{\gamma_{P_S}}\,p$ (see (2)).

Separations and quality shocks. Match quality is hit by shocks at Poisson rate $\lambda_S > 0$. Upon a shock, quality is redrawn i.i.d. from Γ on $[0,1]$ with density $\gamma(\cdot)$. A match is destroyed whenever the post-shock draw is below the reservation cutoff $p_S^*(x) \in (0,1)$; otherwise the match continues at the new quality. In addition, a filled match is destroyed exogenously at Poisson rate $\xi_S \geq 0$. Skilled separations therefore come from the endogenous mechanism, the exogenous hazard ξ_S, or demographic exit at rate ν.

Interpretation of Γ. Γ is the match-quality distribution in the skilled market. We parameterise $\Gamma = \mathrm{Beta}(a_\Gamma, b_\Gamma)$.

OJS (discrete choice). An employed skilled worker chooses whether to search while employed. Searching incurs a flow cost $\sigma_S \geq 0$. If searching, the worker receives outside offers at rate $\kappa_S \equiv \theta_S q_S(\theta_S)$. Each offer is a draw $\tilde{p} \sim \Gamma$. The worker quits to the outside job iff $\tilde{p} > p$.

Demographic turnover. All workers exit at Poisson rate $\nu > 0$.

8.2 Matching and tightness

Let $u_S(x)$ denote the density of skilled unemployed by type, and let $e_S(x,p)$ denote the density of employed trained workers by (x,p). Only workers with $d(x) = 0$ search the skilled market when unemployed; employed skilled workers may search via OJS, captured by $s^*(x,p) \in \{0,1\}$ (defined below). Define the aggregate mass of skilled job seekers as

$$\widetilde{\mathcal{U}}_S \;\equiv\; \underbrace{\int_0^1 (1-d(x))\,u_S(x)\,\mathrm{d}x}_{\text{unemployed S-searchers}} + \underbrace{\int_0^1\int_0^1 s^*(x,p)\,e_S(x,p)\,\mathrm{d}p\,\mathrm{d}x}_{\text{employed S-searchers}}.$$

Let $\mathcal{V}_S$ be the mass of skilled vacancies; tightness is

$$\theta_S = \frac{\mathcal{V}_S}{\widetilde{\mathcal{U}}_S}, \qquad q_S(\theta_S) = \mu_S\,\theta_S^{-\eta_S}, \qquad \kappa_S \equiv \theta_S\,q_S(\theta_S).$$

8.3 Two unemployment values and the discrete search-market choice

Let:

- $U_S(x)$ be the value of skilled unemployment;
- $E_S^0(x,p)$ and $E_S^1(x,p)$ be worker values in a job of quality p without and with OJS;
- $J_S^0(x,p)$ and $J_S^1(x,p)$ be the corresponding firm values;
- V_S be the value of a skilled vacancy.

Define the worker's continuation value in a job of quality p as

$$E_S(x,p) \equiv \max\{E_S^0(x,p),\ E_S^1(x,p)\}, \qquad J_S(x,p) \equiv J_S^{s^*(x,p)}(x,p).$$

Two candidate unemployment values. Conditional on a fixed search-market policy $d(x)$, the unemployment Bellman is

$$(r+\nu)\,U_S(x) \;=\; b_S \;+\; (1-d(x))\,A_S(x) \;+\; d(x)\,A_U(x),$$

with the skilled-market option value

$$A_S(x) \;=\; \kappa_S \int_{p_S^*(x)}^1 \big(E_S(x,\tilde{p}) - U_S(x)\big)\,\mathrm{d}\Gamma(\tilde{p}),$$

and the unskilled-market option value

$$A_U(x) \;=\; f_U\,\big(E_U(x,1) - U_S(x)\big)_+,$$

where $E_U(x,1) = U_U(x) + \beta_U\,S_U(x,1)$ is the worker value of a fresh unskilled match (worker treated as untrained at bargaining). Define the two *candidate* unemployment values:

$$U_S^{(0)}(x) \;\equiv\; \frac{b_S + \kappa_S\beta_S\,\mathcal{I}_S(x)}{r+\nu}, \tag{19}$$

$$U_S^{(1)}(x) \;\equiv\; \frac{b_S + f_U\,E_U(x,1)}{r+\nu+f_U}, \tag{20}$$

where $\mathcal{I}_S(x) \equiv \int_{p_S^*(x)}^1 S_S(x,\tilde p)\,\mathrm{d}\Gamma(\tilde p)$ is the skilled tail integral and we used Nash bargaining ($E_S - U_S = \beta_S S_S$) in A_S. The form of $U_S^{(1)}(x)$ embeds the optimality of *always accepting* an unskilled offer when $d(x)=1$ (the worker would otherwise switch to $d=0$).

Discrete policy. The worker picks the market that delivers the higher unemployment value:

$$d(x) \;=\; \mathbf{1}\{U_S^{(1)}(x) > U_S^{(0)}(x)\}, \qquad U_S(x) \;=\; \max\{U_S^{(0)}(x),\, U_S^{(1)}(x)\}. \tag{21}$$

Under monotonicity (which typically holds in the parameter region of interest), d is a single-cutoff policy in x. Low-ability trained workers, for whom the skilled option value $\kappa_S\beta_S\mathcal{I}_S(x)$ is small, choose $d=1$; high- ability workers choose $d=0$.

8.4 Employed-worker HJBs

Employed worker, no search ($s=0$). With no search, the events are demographic exit ν, quality shocks λ_S, and exogenous separation ξ_S. The worker HJB is

$$\begin{aligned}(r+\nu+\lambda_S+\xi_S)\,E_S^0(x,p) &= w_S^0(x,p) + \xi_S\,U_S(x)\\ &\quad + \lambda_S\left[U_S(x)\,\Gamma(p_S^*(x)) + \int_{p_S^*(x)}^1 E_S(x,p')\,\mathrm{d}\Gamma(p')\right].\end{aligned} \tag{22}$$

Employed worker, search ($s=1$).

$$\begin{aligned}(r+\nu+\lambda_S+\xi_S)\,E_S^1(x,p) &= (w_S^1(x,p) - \sigma_S) + \xi_S\,U_S(x)\\ &\quad + \lambda_S\left[U_S(x)\,\Gamma(p_S^*(x)) + \int_{p_S^*(x)}^1 E_S(x,p')\,\mathrm{d}\Gamma(p')\right]\\ &\quad + \kappa_S\int_{\max\{p,p_S^*(x)\}}^1 \big(E_S(x,\tilde p) - E_S^1(x,p)\big)\,\mathrm{d}\Gamma(\tilde p).\end{aligned} \tag{23}$$

Firm values. A filled match yields the firm nothing on an exogenous separation, so ξ_S enters only the effective discount rate:

$$\begin{aligned}(r+\nu+\lambda_S+\xi_S)\,J_S^0(x,p) &= A\,\gamma_{P_S}\,x^{\gamma_{P_S}}\,p - w_S^0(x,p)\\ &\quad + \lambda_S\int_{p_S^*(x)}^1 J_S(x,p')\,\mathrm{d}\Gamma(p').\end{aligned} \tag{24}$$

$$\begin{aligned}(r+\nu+\lambda_S+\xi_S+\kappa_S(1-\Gamma(p)))\,J_S^1(x,p) &= A\,\gamma_{P_S}\,x^{\gamma_{P_S}}\,p - w_S^1(x,p)\\ &\quad + \lambda_S\int_{p_S^*(x)}^1 J_S(x,p')\,\mathrm{d}\Gamma(p').\end{aligned} \tag{25}$$

Crucially, $U_S(x)$ in (22)–(25) is the *discrete-choice max* from (21). No other modification of the employed-worker HJBs is needed: the cross-market mechanism enters the skilled block solely through the $U_S(x)$ that the surplus block consumes.

8.5 Policy rules: endogenous separation and OJS cutoff

Define overall surplus $S_S(x,p) \equiv \max\{S_S^0(x,p), S_S^1(x,p)\}$. Endogenous separation is characterised by a cutoff $p_S^*(x)$ satisfying

$$S_S(x, p_S^*(x)) = 0, \qquad S_S(x,p) \geq 0 \iff p \geq p_S^*(x). \tag{26}$$

The worker searches on the job iff $s^*(x,p) = \mathbf{1}\{E_S^1(x,p) \geq E_S^0(x,p)\}$, with cutoff $p_S^{\mathrm{oj}}(x) \geq p_S^*(x)$ such that

$$E_S^1(x,p) \geq E_S^0(x,p) \iff p < p_S^{\mathrm{oj}}(x), \quad E_S^1(x, p_S^{\mathrm{oj}}(x)) = E_S^0(x, p_S^{\mathrm{oj}}(x)). \tag{27}$$

8.6 Surplus and Wages

Because $E_S(x,p) = U_S(x) + \beta_S S_S(x,p)$ and $J_S(x,p) = (1-\beta_S) S_S(x,p)$ under the optimal regime, summing the worker and firm HJBs and subtracting the unemployment HJB eliminates wages. The added $\xi_S U_S$ terms cancel against $-(r+\nu+\xi_S) U_S$, so ξ_S survives only in the effective destruction rate. Substituting $U_S(x)$ from (21) yields:

Surplus with no search ($s = 0$).

$$(r+\nu+\lambda_S+\xi_S)\, S_S^0(x,p) = A\,\gamma_{P_S}\, x^{\gamma_{P_S}}\, p - (r+\nu)\, U_S(x) + \lambda_S\, \mathcal{I}_S(x). \tag{28}$$

Surplus with OJS ($s = 1$).

$$\begin{aligned}
&\big(r+\nu+\lambda_S+\xi_S+\kappa_S(1-\Gamma(p))\big)\, S_S^1(x,p) \\
&\qquad = A\,\gamma_{P_S}\, x^{\gamma_{P_S}}\, p - (r+\nu)\, U_S(x) - \sigma_S + \lambda_S\, \mathcal{I}_S(x) + \kappa_S \beta_S \int_p^1 S_S(x,p')\, \mathrm{d}\Gamma(p'). \qquad (29)
\end{aligned}$$

8.7 Wage schedules

$$w_S^0(x,p) = \beta_S\, A\,\gamma_{P_S}\, x^{\gamma_{P_S}}\, p + (1-\beta_S)(r+\nu)\, U_S(x), \tag{30}$$

$$\begin{aligned}
w_S^1(x,p) &= \beta_S\, A\,\gamma_{P_S}\, x^{\gamma_{P_S}}\, p + (1-\beta_S)(r+\nu)\, U_S(x) + (1-\beta_S)\,\sigma_S \\
&\qquad - \beta_S(1-\beta_S)\,\kappa_S \int_p^1 S_S(x,p')\, \mathrm{d}\Gamma(p'). \qquad (31)
\end{aligned}$$

The wage schedules are unchanged in form: ξ_S enters wages only through its effect on the surplus S_S and the unemployment value U_S, and A enters only through the output term. The OJS wage premium is the difference of the two schedules:

$$w_S^1(x,p) - w_S^0(x,p) = (1-\beta_S)\,\sigma_S - \beta_S(1-\beta_S)\,\kappa_S \int_p^1 S_S(x,p')\, \mathrm{d}\Gamma(p') \;\leq\; 0, \tag{32}$$

where the inequality holds on the OJS region $p < p_S^{\mathrm{oj}}(x)$ because optimality of search there implies $\beta_S\, \kappa_S \int_p^1 S_S\, \mathrm{d}\Gamma \geq \sigma_S$ (the Pissarides condition).

8.8 Vacancies, poaching, and free entry

Posting a skilled vacancy costs $k_S > 0$ per unit time. Define the density of skilled job seekers (unemployed S-searchers plus employed S-searchers):

$$\widetilde{u}_S(x) \;\equiv\; (1-d(x))\, u_S(x) + \int_0^1 s^*(x,p)\, e_S(x,p)\, \mathrm{d}p, \qquad \widetilde{\mathcal{U}}_S = \int_0^1 \widetilde{u}_S(x)\, \mathrm{d}x.$$

Workers with $d(x) = 1$ are searching in the U-market, not in the S-market, and so contribute zero to the skilled seeker pool. Conditional on a contact, the type is drawn from $\widetilde{u}_S$, and the expected firm surplus is

$$\begin{aligned}\mathcal{J}_S \;\equiv\; \frac{1}{\widetilde{\mathcal{U}}_S}\bigg[&\int_0^1 (1-d(x))\, u_S(x) \int_{p_S^*(x)}^1 J_S(x,\tilde{p})\, \mathrm{d}\Gamma(\tilde{p})\, \mathrm{d}x \\ &+ \int_0^1\!\!\int_0^1 s^*(x,p)\, e_S(x,p) \int_{\max\{p,p_S^*(x)\}}^1 J_S(x,\tilde{p})\, \mathrm{d}\Gamma(\tilde{p})\, \mathrm{d}p\, \mathrm{d}x\bigg].\end{aligned} \tag{33}$$

Because J_S already carries ξ_S in its effective discount rate, free entry needs no separate ξ_S term. Free entry implies $V_S = 0$ and

$$q_S(\theta_S)\, \mathcal{J}_S = k_S. \tag{34}$$

8.9 Stationary cross-sectional distributions (trained workers)

Accounting within the trained population:

$$m_S(x) = u_S(x) + \int_{p_S^*(x)}^1 e_S(x,p)\, \mathrm{d}p.$$

The endogenous destruction hazard, the total destruction hazard (including ξ_S), and the OJS quit hazard are

$$\delta_S^{\mathrm{end}}(x) \equiv \lambda_S\, \Gamma(p_S^*(x)), \qquad \delta_S^{\mathrm{tot}}(x) \equiv \xi_S + \lambda_S\, \Gamma(p_S^*(x)), \qquad \delta_S^{\mathrm{jj}}(x,p) \equiv s^*(x,p)\, \kappa_S\, (1-\Gamma(p)).$$

Unemployment balance (modified). For each x, the outflow from u_S branches on the search-market choice; the inflow from employment carries the total destruction hazard $\xi_S + \delta_S^{\mathrm{end}}(x)$:

$$\Big(\nu + (1-d(x))\, \kappa_S(1-\Gamma(p_S^*(x))) + d(x)\, f_U\Big)\, u_S(x) = \phi\, t(x) + \int_{p_S^*(x)}^1 (\xi_S + \delta_S^{\mathrm{end}}(x))\, e_S(x,p)\, \mathrm{d}p. \tag{35}$$

For $d(x) = 0$ types this is the baseline balance. For $d(x) = 1$ types the matching outflow $\kappa_S(1-\Gamma(p_S^*(x)))$ is replaced by f_U (the worker leaves the segment to take an unskilled job).

Employment-by-quality balance. For $d(x) = 0$ types, the standard balance applies, with ξ_S added to the per-match outflow:

$$\begin{aligned}&\Big(\nu + \lambda_S + \xi_S + \delta_S^{\mathrm{jj}}(x,p)\Big)\, e_S(x,p) \\ &\quad= \kappa_S\, u_S(x)\, \gamma(p) + \lambda_S\, \gamma(p) \int_{p_S^*(x)}^1 e_S(x,p')\, \mathrm{d}p' + \kappa_S\, \gamma(p) \int_{p_S^*(x)}^p s^*(x,p')\, e_S(x,p')\, \mathrm{d}p'.\end{aligned} \tag{36}$$

For $d(x) = 1$ types, there is no inflow into skilled employment because the worker is searching the U-market, so in stationarity

$$e_S(x,p) = 0 \quad \text{for all } p, \text{ whenever } d(x) = 1. \tag{37}$$

Combined with (35), this gives the closed form $u_S(x) = \phi\, t(x)/(\nu + f_U)$ on the $d = 1$ region, consistent with (18).

Remark 2 (Implementation)**.** *On a grid $p_1 < \cdots < p_{N_p}$, equation (36) becomes a linear system in the unknown vector $\{e_S(x,p_i)\}_{i=1}^{N_p}$ for each x with $d(x) = 0$; the only change relative to the $\xi_S = 0$ case is the $+\xi_S$ term in the diagonal outflow. The density $u_S(x)$ is then closed by the mass identity $m_S(x) = u_S(x) + \int e_S\, \mathrm{d}p$, so the extra exogenous outflow propagates automatically. For x with $d(x) = 1$ the system collapses to $e_S(x,p_i) = 0$, $u_S(x) = \phi\, t(x)/(\nu + f_U)$.*

9 Observed wages and measurement error

The structural wage schedules (12), (30)–(31) describe the wage paid in equilibrium. Survey wages, however, are measured with error. We model this as multiplicative log-normal noise on the observed wage, independent across workers and of all model objects:

$$\log w_j^{\text{obs}}(x,p) \;=\; \log w_j(x,p) \;+\; \sigma_{w,j}\,\varepsilon, \qquad \varepsilon \sim N(0,1), \qquad j \in \{U,S\}. \tag{38}$$

This layer is applied *after* the equilibrium is solved; it changes no value function, policy, density, or tightness. It affects only the wage moments that the SMM targets, and only in the following way. Let $\mu_{\log}, \sigma^2_{\log}, \mu_{3,\log}$ denote the structural mean, variance, and third central moment of the log wage in sector j (across the employed (x,p) distribution). Then the observed counterparts are

$$\mathbb{E}[\log w_j^{\text{obs}}] = \mu_{\log}, \qquad \mathrm{Var}[\log w_j^{\text{obs}}] = \sigma^2_{\log} + \sigma^2_{w,j}, \qquad \mu_3[\log w_j^{\text{obs}}] = \mu_{3,\log},$$

because the noise is mean-zero and symmetric. The observed α-quantile is the convolution of the structural log-wage distribution with $N(0,\sigma^2_{w,j})$. Hence:

- `mean_wage_j`, `wage_premium`, `emp_cm3_j` are invariant to $\sigma_{w,j}$;
- `emp_var_j` is inflated by $\sigma^2_{w,j}$;
- `p25_wage_j`, `p50_wage_j` are the $\sigma_{w,j}$-convolved quantiles.

$\sigma_{w,j}$ therefore absorbs the residual cross-sectional log-wage variance that the search model cannot generate at plausible bargaining weights (the Hornstein–Krusell–Violante dispersion gap), and is identified off `emp_var_j` net of the structural dispersion pinned by the transition and skewness moments. Setting $\sigma_{w,j} = 0$ returns the structural wages.

10 Skilled-block equilibrium and nested fixed-point algorithm

Take the trained population density $m_S(\cdot)$, the unskilled contact rate f_U, and the unskilled frontier worker value $E_U(\cdot,1)$ as given from the unskilled block. A *skilled-block equilibrium* is a collection

$$\{\theta_S,\ U_S(\cdot),\ E_S^0(\cdot,\cdot),\ E_S^1(\cdot,\cdot),\ J_S^0(\cdot,\cdot),\ J_S^1(\cdot,\cdot),\ w_S^0(\cdot,\cdot),\ w_S^1(\cdot,\cdot),$$
$$d(\cdot),\ p_S^*(\cdot),\ p_S^{\text{oj}}(\cdot),\ s^*(\cdot,\cdot),\ u_S(\cdot),\ e_S(\cdot,\cdot)\}$$

such that:

(S1) **Worker values.** Unemployment satisfies the discrete-choice envelope (21) with branches (19)–(20). Employed values satisfy (22)–(23).
(S2) **Firm values.** Filled-job values satisfy (24)–(25).
(S3) **OJS policy.** $s^*(x,p)$ satisfies (27).
(S4) **Endogenous separation.** The reservation cutoff $p_S^*(x)$ satisfies (26).
(S5) **Wage determination.** Wages satisfy Nash bargaining (equivalently (30)–(31)).
(S6) **Tightness and free entry.** Tightness is defined on the skilled seeker pool $\widetilde{\mathcal{U}}_S$ (excluding $d=1$ types) and free entry holds: $q_S(\theta_S)\,\mathcal{J}_S = k_S$.
(S7) **Stationary distributions.** Given $(\theta_S, p_S^*, p_S^{\text{oj}}, s^*, d)$, the densities (u_S, e_S) satisfy (35)–(37) and the accounting identity $m_S(x) = u_S(x) + \int_{p_S^*(x)}^1 e_S(x,p)\,\mathrm{d}p$.

10.1 Nested fixed-point algorithm (skilled side)

The skilled block is organised as a nested fixed point.

Outer-loop unknowns. Iterate on $(\theta_S^{(n)},\ p_S^{*,(n)}(\cdot),\ p_S^{\text{oj},(n)}(\cdot))$, with f_U, $E_U(\cdot,1)$, $m_S(\cdot)$ held fixed from the latest unskilled solve.

Step 1. Initialise (outer loop). Choose $\theta_S^{(0)} > 0$ and feasible cutoffs $p_S^{*,(0)}(x)$, $p_S^{\text{oj},(0)}(x)$.

Step 2. Inner loop: surplus iteration. Holding $(\theta_S^{(n)}, p_S^{*,(n)}, p_S^{\text{oj},(n)})$ fixed, iterate on (U_S, S_S^0, S_S^1, d) until convergence. Let $\kappa_S = \theta_S^{(n)} q_S(\theta_S^{(n)})$ throughout. For each x and each pass m:

- **Step 2.1. Tail integral.** Let j_0 be the first grid index with $p_{j_0} \geq p_S^{*,(n)}(x)$. Accumulate

$$\mathcal{I}^{(m)}(x) = \int_{p_S^{*,(n)}(x)}^{1} S_S^{\max,(m)}(x, p')\, \mathrm{d}\Gamma(p'), \quad S_S^{\max,(m)} = \max\{S_S^{0,(m)},\, S_S^{1,(m)}\}.$$

- **Step 2.2. Two candidate U_S branches.** Compute

$$U_S^{(0)}(x) \;=\; \frac{b_S + \kappa_S \beta_S \mathcal{I}^{(m)}(x)}{r+\nu}, \qquad U_S^{(1)}(x) \;=\; \frac{b_S + f_U\, E_U(x,1)}{r+\nu+f_U},$$

  and set

$$d^{(m+1)}(x) \;=\; \mathbf{1}\{U_S^{(1)}(x) > U_S^{(0)}(x)\}, \qquad U_S^{(m+1)}(x) \;=\; \max\{U_S^{(0)}(x),\, U_S^{(1)}(x)\}.$$

- **Step 2.3. Surplus on p-grid.** Let base $\equiv r + \nu + \lambda_S + \xi_S$. For each p_j with $j \geq j_0$:

$$S_S^{0,(m+1)}(x, p_j) = \frac{A\, \gamma_{P_S}\, x^{\gamma_{P_S}}\, p_j - (r+\nu)\, U_S^{(m+1)} + \lambda_S\, \mathcal{I}^{(m)}}{\text{base}},$$
$$S_S^{1,(m+1)}(x, p_j) = \frac{A\, \gamma_{P_S}\, x^{\gamma_{P_S}}\, p_j - (r+\nu)\, U_S^{(m+1)} - \sigma_S + \lambda_S\, \mathcal{I}^{(m)} + \kappa_S\, \beta_S \int_{p_j}^{1} S_S^{\max,(m)}\, \mathrm{d}\Gamma}{\text{base} + \kappa_S(1-\Gamma(p_j))}.$$

  For $j < j_0$, $S_S^{0,(m+1)} = S_S^{1,(m+1)} = 0$.
- **Step 2.4. Nash split.** $E_S^{s,(m+1)} = U_S^{(m+1)} + \beta_S S_S^{s,(m+1)}$, $J_S^{s,(m+1)} = (1-\beta_S) S_S^{s,(m+1)}$, $s \in \{0,1\}$.

Stop when $\sup_{x,p} \max\{|S_S^{0,(m+1)} - S_S^{0,(m)}|, |S_S^{1,(m+1)} - S_S^{1,(m)}|, |U_S^{(m+1)} - U_S^{(m)}|\} < \text{tol\_inner}$. Note that $d(x)$ is a by-product of the value iteration and requires no separate fixed-point loop; its discrete jumps occur at the boundary of the $U_S^{(0)} = U_S^{(1)}$ crossing.

Step 3. Update cutoffs (outer update). Locate $p_S^{*,(n+1)}(x)$ at the first grid point where $S_S^{\max,(m+1)}(x, p_j) \geq 0$, and $p_S^{\text{oj},(n+1)}(x)$ at the first grid point where $S_S^{1,(m+1)}(x,p_j) - S_S^{0,(m+1)}(x,p_j) \leq 0$, both with damped updates.

Step 4. Stationary distributions (branch on d). For x with $d(x) = 0$: solve the standard flow-balance system (35)–(36) on the discretised p grid (the $+\xi_S$ term enters the diagonal outflow). For x with $d(x) = 1$: use the closed form

$$u_S(x) \;=\; \frac{\phi\, t(x)}{\nu + f_U}, \qquad e_S(x,p) \;=\; 0 \quad \text{for all } p.$$

Compute the seeker density $\tilde{u}_S(x) = (1 - d(x))\, u_S(x) + \int_0^1 s^*(x,p)\, e_S(x,p)\, \mathrm{d}p$.

Step 5. Update tightness θ_S by free entry. Compute $\mathcal{J}_S$ from (33) using (u_S, e_S, s^*, d) and the converged $J_S(x,p)$. Then

$$q_S(\theta_S^{(n+1)}) \;=\; \frac{k_S}{\mathcal{J}_S}, \qquad \theta_S^{(n+1)} = (\mu_S / q_S(\theta_S^{(n+1)}))^{1/\eta_S}.$$

Apply Anderson acceleration to reduce fluctuations.

Step 6. Convergence (outer loop). Stop if

$$\max\Big\{|\theta_S^{(n+1)} - \theta_S^{(n)}|,\ \|p_S^{*,(n+1)} - p_S^{*,(n)}\|_\infty, \|p_S^{\mathrm{oj},(n+1)} - p_S^{\mathrm{oj},(n)}\|_\infty\Big\} < \mathrm{tol}.$$

Otherwise set $n \leftarrow n+1$ and return to Step 2.

Link to the unskilled block. On convergence, the skilled block exposes $U_S(\cdot)$ and the product $d(\cdot)\,u_S(\cdot)$ to the unskilled block; the unskilled block exposes f_U, $E_U(\cdot,1)$, and $m_S(\cdot)$ to the skilled block.

11 Full model equilibrium and global algorithm

A *full equilibrium* consists of unskilled-block and skilled-block objects such that:

(F1) The unskilled block satisfies conditions (E1)–(E5) given the skilled unemployment value $U_S(\cdot)$ and the cross-market seeker contribution $d(\cdot)\,u_S(\cdot)$.

(F2) The skilled block satisfies conditions (S1)–(S7) given the trained density

$$m_S(x) = \frac{\phi\, t(x)}{\nu + d(x)\, f_U},$$

the unskilled contact rate f_U, and the frontier worker value $E_U(\cdot,1)$.

(F3) The link variables are consistent: $U_S(\cdot)$, $m_S(\cdot)$, $E_U(\cdot,1)$, f_U, and $d(\cdot)\,u_S(\cdot)$ are unchanged across the last global iteration.

11.1 Global fixed-point algorithm

The full model is solved via an outer fixed point on the link variables. Only $(U_S(\cdot), m_S(\cdot))$ are accelerated by Anderson; the remaining link objects are derived deterministically within each pass.

Step 1. Initialise. Choose initial guesses $U_S^{(0)}(\cdot)$ and $m_S^{(0)}(\cdot)$; set $d^{(0)}(\cdot)u_S^{(0)}(\cdot) \equiv 0$ (degenerates to the baseline unskilled solve on the first pass).

Step 2. Solve the unskilled block. Given $U_S^{(n)}(\cdot)$ and the carried $d^{(n)}(\cdot)\,u_S^{(n)}(\cdot)$, solve the unskilled equilibrium as in Section 7. Form

$$S_U^{(n)}(\cdot,1) = J_U^{(n)}(\cdot,1)/(1-\beta_U), \qquad E_U^{(n)}(\cdot,1) = U_U^{(n)}(\cdot) + \beta_U\, S_U^{(n)}(\cdot,1), \qquad f_U^{(n)} = \theta_U^{(n)} q_U(\theta_U^{(n)}).$$

Step 3. Solve the skilled block. Given $m_S^{(n)}$, $f_U^{(n)}$, $E_U^{(n)}(\cdot,1)$, solve the skilled equilibrium of Section 10. Obtain $U_S^{\mathrm{new}}(\cdot)$, $d^{\mathrm{new}}(\cdot)$, and $u_S^{\mathrm{new}}(\cdot)$.

Step 4. Update link variables. Compute

$$m_S^{\mathrm{new}}(x) = \frac{\phi\, t^{(n)}(x)}{\nu + d^{\mathrm{new}}(x)\, f_U^{(n)}}.$$

Apply Anderson($m=1$) to the joint state (U_S, m_S) to obtain $(U_S^{(n+1)}, m_S^{(n+1)})$. Write back to the unskilled cache:

$$d^{(n+1)}(\cdot)\, u_S^{(n+1)}(\cdot) \;\leftarrow\; d^{\mathrm{new}}(\cdot)\, u_S^{\mathrm{new}}(\cdot).$$

Step 5. Convergence. Stop if

$$\max\{\|U_S^{(n+1)} - U_S^{(n)}\|_\infty,\ \|m_S^{(n+1)} - m_S^{(n)}\|_\infty\} < \text{tol}.$$

Otherwise set $n \leftarrow n+1$ and return to Step 2.

12 Transition dynamics: evolution of densities after a parameter change

Consider an unexpected, permanent change in some parameters at calendar time T. Where the distinction matters, we denote pre-switch quantities with superscript "pre" and post-switch quantities with "post". For $t > T$, all hazards and policies are those implied by the post-switch parameters.

Equilibrium object. A transition equilibrium is a collection of time paths for value functions, policies (including $d(\cdot, t)$), market tightness, and cross-sectional distributions such that:

1. Value functions solve time-dependent HJB equations given $(\theta_U(t), \theta_S(t))$;
2. Policies $(\tau, p_U^*, p_S^*, p_S^{\text{oj}}, d)$ are optimal pointwise;
3. Distributions solve the laws of motion forward in time;
4. Market tightness satisfies free entry at each date;
5. Initial distributions match the pre-switch steady state and the economy converges to the post-switch steady state.

12.1 Time-dependent objects and hazards

All objects are time-dependent for $t > T$. Let $u_U(x,t)$, $t(x,t)$, $u_S(x,t)$, $e_S(x,p,t)$ denote the densities, and $\theta_U(t)$, $\theta_S(t)$ the tightnesses, with

$$f_U(t) = \theta_U(t) q_U(\theta_U(t)), \qquad f_S(t) = \theta_S(t) q_S(\theta_S(t)).$$

Policies are time-dependent:

$$\tau(x,t), \quad p_U^*(x,t), \quad p_S^*(x,t), \quad p_S^{\text{oj}}(x,t), \quad d(x,t).$$

Endogenous destruction hazards (and the total skilled hazard including ξ_S):

$$\delta_U(x,t) = \lambda_U\, G(p_U^*(x,t)), \qquad \delta_S^{\text{end}}(x,t) = \lambda_S\, \Gamma(p_S^*(x,t)), \qquad \delta_S^{\text{tot}}(x,t) = \xi_S + \lambda_S\, \Gamma(p_S^*(x,t)).$$

12.2 Segment accounting identities

Define

$$m_U(x,t) \equiv u_U(x,t) + t(x,t) + e_U(x,t), \qquad m_S(x,t) \equiv u_S(x,t) + e_S^{\text{tot}}(x,t),$$

where $e_S^{\text{tot}}(x,t) \equiv \int_{p_S^*(x,t)}^1 e_S(x,p,t)\, \mathrm{d}p$.

12.3 Laws of motion: unskilled segment

Training stock.

$$\partial_t\, t(x,t) = \tau(x,t)\, u_U(x,t) - (\phi + \nu)\, t(x,t). \tag{39}$$

Untrained unemployment. The inflow from cross-market separations enters through the residual e_U on the destruction term (since skilled-detoured employed are pooled into e_U). The balance retains its baseline form:

$$\partial_t\, u_U(x,t) = \nu\, \ell(x) + \delta_U(x,t)\, e_U(x,t) - \big(f_U(t) + \tau(x,t) + \nu\big)\, u_U(x,t). \tag{40}$$

Untrained-segment mass by type. The population identity $m_U(x,t) = \ell(x) - m_S(x,t)$ gives $\partial_t m_U(x,t) = -\partial_t m_S(x,t)$, so the substantive ODE is (42) below.

12.4 Laws of motion: skilled segment

Skilled unemployment (modified). The inflow from employment carries the total destruction hazard $\xi_S + \delta_S^{\text{end}}(x,t)$:

$$\begin{aligned}\partial_t\, u_S(x,t) \;=\; &\phi\, t(x,t) + \left(\xi_S + \delta_S^{\text{end}}(x,t)\right) e_S^{\text{tot}}(x,t) \\ &- \Big(\nu + (1-d(x,t))\, f_S(t)\, \left(1-\Gamma(p_S^*(x,t))\right) + d(x,t)\, f_U(t)\Big)\, u_S(x,t).\end{aligned} \tag{41}$$

Trained-segment mass by type (unchanged).

$$\partial_t\, m_S(x,t) \;=\; \phi\, t(x,t) - \nu\, m_S(x,t) - d(x,t)\, f_U(t)\, u_S(x,t). \tag{42}$$

The extra outflow $d(x,t)\, f_U(t)\, u_S(x,t)$ moves mass from the skilled segment to the unskilled segment; it does not affect the total population identity because the receiving workers are absorbed into e_U on the unskilled side. Exogenous separation ξ_S does not appear here: it reshuffles mass between e_S and u_S *within* the skilled segment and so leaves m_S unchanged.

12.5 Aggregate mass dynamics

Integrating (42) over x:

$$\dot{M}_S(t) = \phi\, \mathcal{T}(t) - \nu\, M_S(t) - f_U(t) \int_0^1 d(x,t)\, u_S(x,t)\, \mathrm{d}x,$$

where $\mathcal{T}(t) \equiv \int_0^1 t(x,t)\, \mathrm{d}x$. The total-population invariance $M_U(t) + M_S(t) = \text{const}$ is preserved because the cross-market outflow from m_S reappears as an inflow to m_U via e_U, leaving the sum unchanged.

12.6 Initial condition at the switch

Let $(u_U^{\text{pre}}(x),\, t^{\text{pre}}(x),\, u_S^{\text{pre}}(x),\, m_S^{\text{pre}}(x))$ denote the pre-switch stationary densities. At $t = T^+$,

$$u_U(x,T^+) = u_U^{\text{pre}}(x), \quad t(x,T^+) = t^{\text{pre}}(x), \quad u_S(x,T^+) = u_S^{\text{pre}}(x), \quad m_S(x,T^+) = m_S^{\text{pre}}(x).$$

In a tranquil pre-switch regime we typically have $d^{\text{pre}}(x) \equiv 0$, so the initial condition coincides with the baseline pre-switch state.

12.7 Market tightness along the transition

Unskilled market (modified).

$$q_U(\theta_U(t)) \;=\; \frac{k_U\, \widetilde{\mathcal{U}}_U(t)}{\int_0^1 J_U(x,1,t)\, \left(u_U(x,t) + d(x,t)\, u_S(x,t)\right) \mathrm{d}x},$$

where $\widetilde{\mathcal{U}}_U(t) = \int_0^1 (u_U + d\, u_S)(x,t)\, \mathrm{d}x$.

Skilled market.

$$q_S(\theta_S(t)) = \frac{k_S}{\mathcal{J}_S(t)},$$

where $\mathcal{J}_S(t)$ uses the time-t seeker density excluding $d(\cdot,t) = 1$ types.

12.8 Transition mapping

At each t:

$$(u_U, t, u_S, e_S)_t \;\Rightarrow\; (\theta_U, \theta_S)_t \;\Rightarrow\; \text{value functions} \;\Rightarrow\; (\tau, p_U^*, p_S^*, p_S^{\mathrm{oj}}, d)_t \;\Rightarrow\; \partial_t(u_U, t, u_S, e_S).$$

12.9 Exact transition equilibrium: numerical algorithm

The transition path is a forward–backward fixed point.

Step 0. Boundary objects. Compute the pre- and post-switch stationary equilibria (both under the DC mechanism).

Step 1. Initial guess. Initial paths $\{\theta_U^{(0)}(t_n), \theta_S^{(0)}(t_n)\}_{n=0}^N$.

Step 2. Backward pass. Given $(\theta_U^{(k)}, \theta_S^{(k)})$, solve the HJB system backward from t_N to t_0. At each t_n compute the two U_S branches (19)–(20), set $d(x, t_n)$ by (21), and propagate the surplus equations.

Step 3. Forward pass. Given the policy paths $(\tau, p_U^*, p_S^*, p_S^{\mathrm{oj}}, d)$, integrate the laws of motion (39), (40), (41) from t_0 to t_N.

Step 4. Update tightness paths. Recompute $\theta_U^{\mathrm{new}}(t_n), \theta_S^{\mathrm{new}}(t_n)$ from free entry with the augmented unskilled seeker pool and the seeker-corrected skilled pool.

Step 5. Convergence. Iterate Steps 2–4 until

$$\max_n \Big(|\theta_U^{(k+1)}(t_n) - \theta_U^{(k)}(t_n)|,\ |\theta_S^{(k+1)}(t_n) - \theta_S^{(k)}(t_n)| \Big) < \mathrm{tol}.$$

13 Model Properties

This section collects key qualitative properties of the model that shape the economic interpretation of the estimation results and the transition-dynamics exercises.

13.1 Training cutoff and the education margin

Proposition 1 (Training cutoff)**.** *Under the maintained functional forms ($c'(x) < 0$, U_S and T continuous in x), the net training gain*

$$\Delta_T(x) \;=\; -\,c(x) + T(x) - U_U^{\mathrm{search}}(x)$$

is increasing in x. Hence the optimal training policy is a cutoff rule: $\tau(x) = \mathbf{1}\{x > \bar{x}\}$, $\Delta_T(\bar{x}) = 0$. Workers with innate type $x > \bar{x}$ train upon entering unskilled unemployment; those with $x \le \bar{x}$ search in the unskilled market.

Intuition. Higher-ability workers benefit from training through two channels: (i) the direct cost $c(x) = e^c\,(1-x)\,e^{-x}$ is decreasing in x, and (ii) the return to skilled-market access, captured by $T(x) = (b_T + \phi U_S(x))/(r + \phi + \nu)$, is increasing in x because the skilled-side option value $\kappa_S \beta_S \mathcal{I}_S(x)$ rises with ability. Under the DC formulation, $U_S(x) = \max\{U_S^{(0)}(x), U_S^{(1)}(x)\}$ sets a *floor* on the training return at $U_S^{(1)}(x)$, so the relative payoff to training is muted whenever the cross-market option binds.

13.2 Cross-market search cutoff

Proposition 2 (Search-market cutoff)**.** *Suppose $\mathcal{I}_S(x)$ is increasing in x (which holds under $\gamma_{P_S} > 1$ and an interior $p^*_S(x)$). Then $U^{(1)}_S(x) - U^{(0)}_S(x)$ is decreasing in x, and there exists $\underline{x} \in [0,1]$ such that $d(x) = 1$ for $x \le \underline{x}$ and $d(x) = 0$ for $x > \underline{x}$, with*

$$\frac{b_S + f_U\, E_U(\underline{x}, 1)}{r + \nu + f_U} = \frac{b_S + \kappa_S \beta_S\, \mathcal{I}_S(\underline{x})}{r + \nu}.$$

Intuition. Low-x trained workers face small skilled-side option values (both $\mathcal{I}_S(x)$ and the skilled match surplus depend on $x^{\gamma_{P_S}}$). Their best outside option is to redirect search to the unskilled market, where the production technology is linear in x rather than convex. High-x workers retain strong incentives to remain in the skilled search pool. *Comparative statics.* $\underline{x}$ moves with the same primitives as $\bar{x}$ but with opposite signs in two key channels. A fall in γ_{P_S} (crisis-style skilled productivity loss) lowers $\mathcal{I}_S(x)$ across the board, raising $\underline{x}$ — *more* types flip to unskilled search. Conversely, a rise in f_U (recovery on the unskilled side) also raises $\underline{x}$. A fall in f_U pushes $\underline{x}$ down to zero, restoring the baseline no-cross-market equilibrium.

13.3 Separation thresholds

Proposition 3 (Unskilled separation cutoff)**.** *For each type x, there exists a unique reservation match quality $p^*(x) \in (0,1)$ satisfying $J_U(x, p^*) = 0$:*

$$p^*(x) = \frac{(r+\nu)\, U_U(x) - \lambda_U \int_{p^*}^{1} S_U(x, p')\, \mathrm{d}G(p')}{A\, P_U\, x}.$$

The cutoff is decreasing in x.

Proposition 4 (Skilled separation and OJS cutoffs)**.** *For each type x:*

1. *There exists a separation cutoff $p^*_S(x)$ satisfying $S_S(x, p^*_S) = 0$. Matches with $p < p^*_S$ are destroyed upon a quality shock.*
2. *There exists an OJS cutoff $p^{\mathrm{oj}}_S(x) \ge p^*_S(x)$ satisfying $E^1_S(x, p^{\mathrm{oj}}_S) = E^0_S(x, p^{\mathrm{oj}}_S)$.*

Both cutoffs are decreasing in x. For types with $d(x) = 1$ the $p^*_S(x)$ and $p^{\mathrm{oj}}_S(x)$ retain their definitions but are not realised in stationarity because $e_S(x,p) = 0$.

In addition to the endogenous channel, a filled skilled match is destroyed at the exogenous rate ξ_S, so the realised skilled separation hazard by type is $\xi_S + \lambda_S\, \Gamma(p^*_S(x))$. This is the empirical object matched to the data separation rate; the endogenous channel alone cannot generate it once $p^*_S(x)$ falls toward zero.

13.4 Skilled unemployment duration

In the model a skilled-unemployed worker of type x leaves unemployment only by accepting an offer or by demographic exit ν: there is no separate labour-force-exit state, so ν is the only route out of the labour force. The unemployment-exit hazard is therefore constant within a type,

$$\delta_S(x) \;=\; \kappa_S\big(1 - \Gamma(p^*_S(x))\big) \;+\; \nu,$$

the sum of the type-x job-finding rate (offers arrive at κ_S and are accepted with probability $1 - \Gamma(p^*_S(x))$) and the demographic exit ν. An ongoing spell is thus exponential within a type; across the pool the share still unemployed after elapsed duration a is the u_S-weighted mixture survivor

$$\Pr(\text{spell} > a) \;=\; \frac{\int_0^1 u_S(x)\, e^{-\delta_S(x)\, a}\, \mathrm{d}x}{\int_0^1 u_S(x)\, \mathrm{d}x}.$$

Because every type's hazard is constant, all negative duration dependence comes from heterogeneity in $\delta_S(x)$: low-acceptance (low-x) types survive longer and are over-represented in the stock (length-biased sampling), which fattens the long-duration tail. With ν common across types and the mean skilled job-finding hazard pinned separately, the long tail identifies the *dispersion* of $\delta_S(x)$. This is the model counterpart of the skilled long-term-unemployment share (`ltu_share_S`) targeted in estimation, evaluated at $a^* = 27\cdot 12/52 \approx 6.23$ months; the data-side construction is given in the companion Data-and-Moments note.

13.5 Wage structure

Wage schedules (12), (30), and (31) are unchanged in form (up to the common factor A on the output term). The within-group dispersion, skill premium, OJS premium, and bargaining-variation properties hold; the aggregate scale A multiplies the level of all wages without altering their shape, and the measurement layer (38) adds dispersion to the observed (but not structural) wage. An additional property follows from the DC mechanism:

5. **Cross-segment composition.** Following a crisis in which $\underline{x}$ rises, the unskilled-employed composition shifts toward higher x (the absorbed ex-trained workers). Aggregate unskilled wages rise through composition alone, even at fixed bargaining and productivity primitives.

13.6 Market tightness and free entry

For the unskilled market:

$$q_U(\theta_U) = \frac{k_U\,(\mathcal{U}_U + \int d\,u_S\,\mathrm{d}x)}{\int_0^1 J_U(x,1)\,(u_U + d\,u_S)(x)\,\mathrm{d}x},$$

and for the skilled market: $q_S(\theta_S)\,\mathcal{J}_S = k_S$, with $\mathcal{J}_S$ defined by (33). Tightness responds to parameter changes through five channels: (i) direct shifts in vacancy costs k_j and in the aggregate scale A and productivity (both raise filled-job values J); (ii) composition effects through changes in the unemployed pool; (iii) shifts in the matching technology μ_j, η_j (Beveridge-curve shift); *(iv) the DC channel*: a shift in $d(\cdot)$ inflates or deflates the unskilled seeker pool and deflates or inflates the skilled seeker pool correspondingly; and (v) shifts in the exogenous skilled hazard ξ_S, which lower J_S through the effective discount rate. Channel (iv) generates a *mechanical* adjustment in (θ_U, θ_S).

13.7 Transition dynamics: adjustment speed

The aggregate skilled share $M_S(t)$ satisfies

$$\dot{M}_S(t) = \phi\,\mathcal{T}(t) - \nu\,M_S(t) - f_U(t)\int_0^1 d(x,t)\,u_S(x,t)\,\mathrm{d}x,$$

which adds a third adjustment channel to demographic turnover and training completion: the *cross-market drain.* When $d(\cdot,t)$ flips from $\equiv 0$ to positive on a measure of types, M_S overshoots its long-run path because u_S drains into the unskilled segment at rate f_U, faster than demographic replacement could remove the same mass. The half-life of M_S adjustment becomes shorter on the crisis-affected region, but recovers to $\ln 2/\nu$ once the cross-market channel closes. *Demand shock vs. matching-efficiency shock.* The estimation is run on four windows (pre-FC, FC, pre-COVID, COVID). In the data, the two crises produce qualitatively different signatures, and the model's regime-specific parameter block maps them onto two distinct regions of the parameter space.

- A **demand shock** (the financial crisis) traces a movement *along* the Beveridge curve. Unemployment rises symmetrically in both segments ($\Delta u_U \approx \Delta u_S$), tightness collapses in both segments (θ_U, θ_S each fall roughly 50%), job-finding rates fall, separation rates rise modestly, and crisis-window vacancies sit close to the pre-FC Beveridge curve in both segments. Within the model, this is captured primarily by a fall in the aggregate scale A (and the relative shifters P_U, γ_{P_S}) together with a rise in k_U, k_S and possibly λ_U, λ_S (and the skilled layoff hazard ξ_S), holding the matching technology (μ_j, η_j) at baseline. The training share rises (the relative return to training increases as P_U falls more than the effective skilled productivity), $\bar{x}$ falls, and the cross-market option does not bind: $\underline{x} \approx 0$.

- A **matching-efficiency / reallocation shock** (COVID) is associated with an *outward* shift of the Beveridge curve in both segments: at the observed unemployment rates, vacancies are far higher than the pre-FC curve predicts (log-deviations of order $+0.9$ on the U side and $+1.4$ on the S side in the crisis window). Tightness θ_U, θ_S *rises* despite the rise in unemployment, job-finding rates barely move, and separation rates rise much more sharply than in FC — especially on the skilled side. The training share *falls*. Within the model this is consistent with (i) a fall in the matching efficiencies μ_U, μ_S (the Beveridge-curve shift proper), (ii) a rise in λ_S and the exogenous skilled layoff hazard ξ_S, and (iii) a fall in γ_{P_S} or a less favourable (a_Γ, b_Γ) that compresses the skilled option value $\mathcal{I}_S(x)$ for low-x trained workers. The cross-market option becomes valuable for those types, $\underline{x}$ rises, and the cross-market drain $d(\cdot)\, f_U\, u_S$ activates — prior trained cohorts exit the skilled segment to take unskilled jobs. The DC channel is what allows the training share to fall in the post-shock equilibrium without a counterfactual rise in $\bar{x}$.

The multi-regime SMM identifies which regime-specific parameters move in each crisis; the transition-dynamics computation traces out the implied path of $(\bar{x}(t), \underline{x}(t), \theta_U(t), \theta_S(t))$ between the pre- and post-shock equilibria. The matching-efficiency channel is the ingredient that lets COVID be modelled as a Beveridge outward shift rather than a movement along the baseline curve.